

L9. Open Quantum Systems and Quantum Errors

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Outline

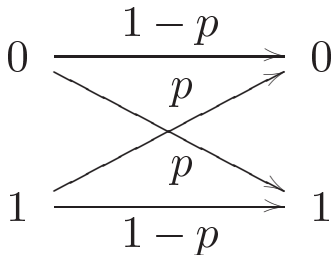
- Errors in a Circuit Model
- Recap of Reduced Density Matrix
- Operator-Sum Representation
- Qubit Decoherence
- *The Master Equation

8.1 Errors in a Circuit Model

- State preparation. Need to couple qubits to other items (cavities, controls, measurement devices) and apply complex protocols to ensure a fast reset.
- Gate imperfection. Imprecise amplitude, phase, or duration of microwave pulses, and qubit frequency fluctuations.
- Measurement errors. Limited integration time, measurement induced decoherence, and amplifier and detector noise.
- Qubit decoherence. Environment induced relaxation, heating, and dephasing, as well as crosstalk, i.e., unwanted interaction between the qubits.

Insights from Classical Noises

- Imagine a bit in a classical computer. The bit starts out in the state 0 or 1, but after a long time the bit may be flipped. We can model this single-stage process by a probability p for the bit to flip, and a probability $1 - p$ for the bit to remain the same.



- The physical origin of the error is that the environment contains electromagnetic fields which cause the bit to flip. Through Maxwell's equations, we can derive the fields in the environment and their interaction with the bit.
- Regardless of the origin, we can simply use the theory of stochastic processes to describe noise in classical systems. A reasonable assumption is to use Markov process in the analysis of multi-stage processes, which means that the process is independent of its past history.

- Therefore, a classical state is described by a vector of probabilities. We only need to know the evolution matrix (or matrix of transition probabilities), which states that

$$p(Y = y) = \sum_x p(Y = y|X = x)p(X = x).$$

- The previous example can be expressed in a matrix form as

$$\begin{bmatrix} p(Y = 0) \\ p(Y = 1) \end{bmatrix} = \begin{bmatrix} 1 - p & p \\ p & 1 - p \end{bmatrix} \begin{bmatrix} p(X = 0) \\ p(X = 1) \end{bmatrix},$$

or, in a short notation,

$$\vec{p}_Y = E\vec{p}_X \equiv \mathcal{E}(\vec{p}_X).$$

8.2 Recap of Reduced Density Matrix

- Recall that the dynamics of a **closed quantum system** are described by a unitary transform. An unitary evolution is a quantum operation:

$$\mathcal{E}_U(\rho) = U\rho U^\dagger.$$

- Also, we can identify projective measurement as a quantum operation as

$$\mathcal{E}_P(\rho) = \sum_k P_k \rho P_k,$$

if the result μ_k of the measurement is lost or unknown.

- Suppose a system (S) and its environment (E) are described, together, by a density matrix ρ .
- If we make a measurement of the system, represented by a linear operator L_S , the averaged measurement result is

$$\begin{aligned}\langle L_S \rangle &\equiv \langle L_S \otimes I_E \rangle = \text{Tr} \rho (L_S \otimes I_E) \\ &= \sum_{i_S, j_E} \langle i_S | \langle j_E | \rho (L_S \otimes I_E) | i_S \rangle | j_E \rangle\end{aligned}$$

- Similar to the definition of trace, we can define **partial trace** of an operator L as

$$\text{Tr}_S L = \sum_i \langle i_S | L | i_S \rangle, \quad \text{Tr}_E L = \sum_j \langle j_E | L | j_E \rangle.$$

- We can use the **reduced density matrix** $\rho_S \equiv \text{Tr}_E \rho = \sum_{j_E} \langle j_E | \rho | j_E \rangle$ to rewrite

$$\begin{aligned} \langle L_S \rangle &= \sum_{i_S, j_E} \langle i_S | \langle j_E | \rho (L_S \otimes I_E) | i_S \rangle | j_E \rangle \\ &= \text{Tr}_S \sum_{j_E} \langle j_E | \rho (L_S \otimes I_E) | j_E \rangle \equiv \text{Tr}_S \rho_S L_S. \end{aligned}$$

- Note that the partial trace is equivalent to a blind measurement of the environment,

$$\rho_S = \text{Tr}_E \sum_{k_E} |k_E\rangle\langle k_E| \rho |k_E\rangle\langle k_E| = \text{Tr}_E [\mathcal{E}_{P_E}(\rho)].$$

8.3 Operator-Sum Representation

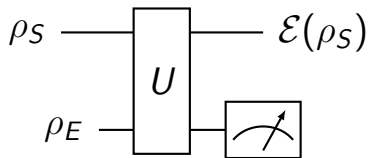
- We now treat an open system S as part of a larger and closed system, which also includes the environment E . For practical interest, we can assume the initial density matrix is separable

$$\rho = \rho_S(0) \otimes \rho_E(0).$$

where $\rho_E(0) = |e_0\rangle\langle e_0|$ is a pure state. Let $|e_k\rangle$ be orthonormal basis for the environmental state space.

- The system and the environment are coupled by a time-independent Hamiltonian H , leading to a unitary time evolution of the total system.

- We then take a partial trace over the environment, which allows us to obtain the reduced state of the system alone.



- As illustrated in the figure, the partial trace can be represented by a measurement in $|e_k\rangle$ basis of the environment (qubits). However, we do not acquire the outcome k from the measurement.

- Put together, with the time evolution operator $U(t) = e^{-iHt/\hbar}$, the evolution of the reduced density matrix (which we care) is

$$\begin{aligned}\rho_S(t) &= \text{Tr}_E \{ U(t) [\rho_S(0) \otimes |e_0\rangle\langle e_0|] U^\dagger(t) \} \\ &= \sum_k \langle e_k | U(t) | e_0 \rangle \rho_S(0) \langle e_0 | U^\dagger(t) | e_k \rangle \\ &= \sum_k E_k \rho_S(0) E_k^\dagger,\end{aligned}$$

where $E_k = \langle e_k | U(t) | e_0 \rangle$ are known as the **Kraus operators** acting on the Hilbert space of the system.

- The Kraus operators satisfy the completeness constraint $\sum_k E_k^\dagger E_k = 1$, if no information about what occurred in the process is obtained by measurement.
- The Kraus operators encode all the available information about the initial state of the environment and about the dynamics of the system-environment coupling. Therefore, the operator-sum representation describes the dynamics of the system without having to explicitly consider properties of the environment.

- Consequently, the reduced density matrix of the system satisfies:
 - $\rho_S(t)$ is Hermitian.
 - $\rho_S(t)$ has unit trace.
 - $\rho_S(t)$ is positive.
- These properties can be compared to the positivity and completeness conditions in the description of classical noise.
- In general, the evolution is not unitary, and $\rho_S(t)$ is in a mixed state after partial trace. There is an arrow of time, i.e., E_k are not invertible.

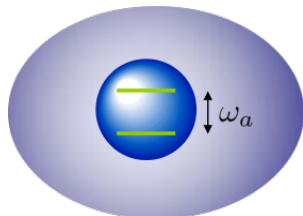
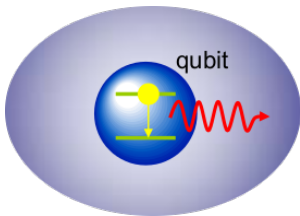
8.4 Qubit Decoherence

- The main source of errors is the impact of noise on a quantum state. As we discussed before, the coupling to environment (or bath) functions as a measurement, but the information is not directly revealed; that is, we study the dynamics of an open quantum system.

Take-Home Messages

- Omnipresent noise can lead to qubit relaxation and dephasing.

- In general, physical qubits are not identical and contains impurities and quasiparticle excitations, hence are subject to various random sources of noise.
- The effects of noise are usually separated into two terms.
 - Relaxation: transition that change the qubit state.
 - Dephasing: modulation of transition frequencies which leads to phase randomization.



Amplitude Damping

- The amplitude-damping channel is a schematic model of the decay of an excited state of a qubit due to spontaneous emission of a photon.
- We denote the qubit ground state by $|g\rangle$ and the excited state of interest by $|e\rangle$. The “environment” is the electromagnetic field, assumed initially to be in its vacuum state $|0\rangle$. After a while, there is a probability p that the excited state has decayed to the ground state and a photon has been emitted, so that the environment has made a transition from the state $|0\rangle$ (“no photon”) to the state $|1\rangle$ (“one photon”).

- The unitary evolution U_{SE} of the qubit and environment is

$$|g\rangle |0\rangle \rightarrow |g\rangle |0\rangle ,$$

$$|e\rangle |0\rangle \rightarrow \sqrt{1-p} |e\rangle |0\rangle + \sqrt{p} |g\rangle |1\rangle .$$

- By evaluating the partial trace over the environment, we find the Kraus operators

$$E_0 = \langle 0|U_{SE}|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} ,$$

$$E_1 = \langle 1|U_{SE}|0\rangle = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix} .$$

- The density matrix evolves as

$$\begin{aligned}
 \rho_S(\delta t) &= E_0 \rho_S(0) E_0^+ + E_1 \rho_S(0) E_1^+ \\
 &= \begin{pmatrix} \rho_{gg} & \sqrt{1-p} \rho_{ge} \\ \sqrt{1-p} \rho_{eg} & (1-p) \rho_{ee} \end{pmatrix} + \begin{pmatrix} p \rho_{ee} & 0 \\ 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} \rho_{gg} + p \rho_{ee} & \sqrt{1-p} \rho_{ge} \\ \sqrt{1-p} \rho_{eg} & (1-p) \rho_{ee} \end{pmatrix}
 \end{aligned}$$

- If we apply the channel n times in succession, we have

$$\rho_{ee} \rightarrow (1-p) \rho_{ee} \rightarrow (1-p)^2 \rho_{ee} \rightarrow \cdots \rightarrow (1-p)^n \rho_{ee}.$$

- Introducing a **relaxation time** T_1 , we can identify the $|e\rangle \rightarrow |g\rangle$ transition probability in given $\delta t \ll T_1$ as $p = \delta t / T_1 \approx 1 - e^{-\delta t / T_1}$. The probability that the excited state persists for time t is $P(t) = (1 - p)^{t/\delta t} \approx e^{-t/T_1}$.
- As $t \rightarrow \infty$, the reduced density matrix approaches

$$\rho_S \xrightarrow{t \rightarrow \infty} \begin{pmatrix} \rho_{gg} + \rho_{ee} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

- So, due to spontaneous emission, the qubit always ends up in its ground state. The coherence between $|g\rangle$ and $|e\rangle$ is also lost (but at half the rate).

Phase Damping

- Phase damping can be modeled as a Brownian motion of the phase, which accumulates random displacements due to the fluctuation in the qubit energy levels.
- The operator-sum description of this quantum process reads

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-q} \end{pmatrix},$$

$$E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{q} \end{pmatrix}.$$

- The density matrix evolves as

$$\begin{aligned}
 \rho_S(\delta t) &= E_0 \rho_S(0) E_0^+ + E_1 \rho_S(0) E_1^+ \\
 &= \begin{pmatrix} \rho_{gg} & \sqrt{1-q} \rho_{ge} \\ \sqrt{1-q} \rho_{eg} & (1-q) \rho_{ee} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & q \rho_{ee} \end{pmatrix} \\
 &= \begin{pmatrix} \rho_{gg} & \sqrt{1-q} \rho_{ge} \\ \sqrt{1-q} \rho_{eg} & \rho_{ee} \end{pmatrix}
 \end{aligned}$$

- Unlike amplitude damping, phase damping leads to a pure decay of the off-diagonal elements of the density matrix. We can identify $\sqrt{1-q}$ as $e^{-\delta t/T_\phi}$ in time interval δt , where T_ϕ is known as the pure **dephasing time**.

- Alternatively, we can write

$$\rho_S(\delta t) = \sqrt{1-q} \begin{pmatrix} \rho_{gg} & \rho_{ge} \\ \rho_{eg} & \rho_{ee} \end{pmatrix} + (1 - \sqrt{1-q}) \begin{pmatrix} \rho_{gg} & 0 \\ 0 & \rho_{ee} \end{pmatrix}.$$

It is as if the qubit were being slightly measured (for small δt , hence $q \ll 1$), continuously in time.

- Eventually, the system collapses into a fixed point, $\rho_\infty = \rho_{gg} |0\rangle\langle 0| + \rho_{ee} |1\rangle\langle 1|$, which is a statistical mixture of $|0\rangle$ and $|1\rangle$, with probabilities that reflect the original state amplitudes.

- If an experiment combines relaxation ($1 - p = e^{-\delta t/T_1}$) and dephasing ($\sqrt{1 - q} = e^{-\delta t/T_\phi}$), the relaxation rate $\gamma = 1/T_1$ remains the same, but the dephasing rate increases to

$$\frac{1}{T_2^*} = \frac{1}{2T_1} + \frac{1}{T_\phi}.$$

- Techniques, such as **spin echo**, exist that can suppress phase damping by introducing spin flips during the experiment, such that quasistatic fluctuations can be cancelled. So qubit relaxation, which scrambles the basis states, is the worse type of decoherence.

8.5 The Master Equation

- In the case of coherent evolution, we describe the dynamics of a quantum system by a differential equation, the Schrödinger equation. The evolution over a finite time interval can be calculated by integrating the equation.
 - The unitary evolution of the corresponding density matrix is known as the Liouville equation (where we set $\hbar = 1$)

$$\dot{\rho} = -i[\mathcal{H}, \rho] \equiv -i(\mathcal{H}\rho - \rho\mathcal{H}).$$

- Now, is it possible to describe decoherence, hence the non-unitary evolution, by a differential equation?

- To a good approximation, yes, but the evolution of the quantum system must be Markovian, i.e., local in time, because $\rho(t + \delta t)$ is completely determined by $\rho(t)$ in a (first-order) differential equation.
- Symbolically, the equation can be written as

$$\dot{\rho} = \mathcal{L}[\rho],$$

where the Lindbladian $\mathcal{L}$ generates non-unitary time evolution in the same sense that a Hamiltonian $\mathcal{H}$ generates unitary evolution.

- We can derive the Lindblad equation from an infinitesimal evolution described by the Kraus sum representation

$$\rho(t + \delta t) = \sum_k E_k(\delta t) \rho(t) E_k^\dagger(\delta t).$$

- For infinitesimal δt , we expect $\rho(t + \delta t) = \rho(t) + \dot{\rho} \delta t$. This means one Kraus operator has the form

$$E_0 = 1 - i\delta t J + O(\delta t^2)$$

with J no longer Hermitian, while the rest have the form

$$E_{k>0} = \sqrt{\delta t} L_k + O(\delta t).$$

- We can identify the normal Hermitian Hamiltonian by

$$\mathcal{H} = \frac{J + J^+}{2},$$

while the Kraus sum completeness constraint requires that

$$K = \frac{J - J^+}{2i} = -\frac{1}{2} \sum_{k>0} L_k^+ L_k.$$

- Taking the limit $\delta t \rightarrow 0$, we obtain

$$\dot{\rho}(t) = -i[\mathcal{H}, \rho(t)] + \sum_{k>0} \left[L_k \rho(t) L_k^+ - \frac{1}{2} L_k^+ L_k \rho(t) - \frac{1}{2} \rho(t) L_k^+ L_k \right].$$

The Lindblad operators L_k describe phenomenological processes, such as spin flips and dephasing.

Summary

- We define partial trace to get rid of the degrees of freedom in unwanted or non-detectable part of a system.
- In the presence of environment, the time evolution can be described by quantum operations on density matrices in the operator-sum representation.
- When a quantum system is entangled with a macroscopic object or environment, the evolution of the quantum state quickly leads to decoherence, and, hence, classical results are recovered.

References

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- I. L. Chuang, Quantum Error Correction, in Quantum Machines: Measurement and Control in Engineered Quantum Systems, edited by M. Devoret, B. Huard, R. Schoelkopf, and L. F. Cugliandolo, Oxford University Press (2014).